

segmentation wonder if there is really a “there there,” as Jonathan Guyton questioned in a 2014 *Journal of Financial Planning* column.

At its core, though, time segmentation simply involves investing differently for retirement spending goals falling at different points in retirement. Fixed-income assets with greater security are generally reserved for earlier retirement expenses, and higher volatility investments with greater growth potential are employed to support later retirement expenses.

The most lucid explanation of time segmentation is provided in the 2004 book *Asset Dedication: How to Grow Wealthy with the Next Generation of Asset Allocation*, written by Stephen J. Huxley and J. Brent Burns. Their description serves as a motivation for this discussion, though I must be clear that I am not specifically replicating their specific approach when I test various time segmentation strategies.

Effectively discussing and simulating time segmentation requires defining it in core terms without some of the bells and whistles that occasionally get included. For example, I will consider two time segments rather than six, use individual bonds for the upfront time segment and a large-capitalization stock fund for the growth segment meant to cover later expenses, and include straightforward automated rules to guide how the upfront time segment is to be extended over time with a rolling bond ladder. This latter point is important, because many versions of time segmentation used in practice may treat the process of extending earlier time segments in an ad hoc fashion, or may even neglect it.

Defining Asset Allocation within Time Segmentation

Building bond ladders to cover retirement spending needs is a relatively understudied topic within the retirement income world. Perhaps the best source for education about the logic of holding individual bonds is provided in the work of Stephen Huxley and J. Brent Burns, who developed the concept of asset dedication. They explain asset allocation differently.

A first consideration for time segmentation is its different treatment of asset allocation. Asset allocation is allowed to fluctuate in a dynamic manner because efforts are not made to maintain a fixed ratio of stocks and bonds. Rather, bond holdings are based on the cost of maintaining a bond ladder with the desired length and income level. Stock holdings consist of what is